Advanced Problem Set

Topic: Trigonometry Flavored Algebra

Handout Instructions

This instructional problem set features rigorous, proof-based subjective tasks highlighting the interplay between algebraic architecture (quadratics, inequalities, and exponents) and functional trigonometric bounds[cite: 5, 29]. Complete all steps systematically.

§1 Single-Answer Subjective Problems

Problem 1

Find the number of pairs (x, y) of real numbers that simultaneously satisfy the following system of equations:

$$|\tan \pi y| + \sin^2 \pi x = 0 \quad \text{and} \quad x^2 + y^2 \leq 2$$

Source: **Math Vanguard Mock Test [cite: 6] (UGA – Q1 [cite: 43, 59])**

Answer Focus: Formulate the condition where both non-negative terms equal zero, then analyze the integer lattices satisfying the boundary disk[cite: 60].

Problem 2

Let a and b be real numbers and let the function f be defined by:

$$f(x) = a \sin x + b\sqrt[3]{x} + 4$$

If $f(\log \log_3 10) = 5$, determine the exact value of $f(\log \log 3)$.
Source: **Math Vanguard Mock Test [cite: 6] (UGA – Q2 [cite: 43, 73])**

Answer Focus: Utilize the properties of logarithms to transform the nested function arguments, recognizing underlying structural symmetry or parity properties[cite: 77, 78].

Problem 3

Consider the real expressions:

$$x = (\sin \alpha)^{\log_b \sin \alpha}$$
$$y = (\cos \alpha)^{\log_b \cos \alpha}$$
$$z = (\sin \alpha)^{\log_b \cos \alpha}$$

where $0 < b < 1$ and $0 < \alpha < \frac{\pi}{4}$. Arrange x, y, and z in increasing order of magnitude.

Source: **Math Vanguard Mock Test [cite: 6] (UGA – Q3 [cite: 43, 82])**

Answer Focus: Apply a base-b logarithm transformation to compare exponential magnitudes via trigonometric monotonicities on the specified domain interval[cite: 88].

Problem 4

Let the logarithmic quantities w, x, y, and z be given by:

$$w = \log_{\sin 1} \cos 1, \quad x = \log_{\sin 1} \tan 1, \quad y = \log_{\cos 1} \sin 1, \quad z = \log_{\cos 1} \tan 1$$

Determine the relative order of w, x, y, and z from smallest to largest.
Source: **Math Vanguard Mock Test [cite: 6] (UGA – Q4 [cite: 43, 101])**

Answer Focus: Use change-of-base rules along with the ordering $0 < \cos 1 < \sin 1 < 1 < \tan 1$ to deduce the structural inequalities[cite: 102, 103, 104, 106].

Problem 5

On the interval $[1, 2]$, the quadratic function $f(x) = x^2 + px + q$ and the function $g(x) = x + \frac{1}{x^2}$ attain the exact same minimum value at the same point. Find the maximum value of $f(x)$ on this interval.
Source: **Math Vanguard Mock Test [cite: 6] (UGA – Q5 [cite: 43, 112])**

Answer Focus: Optimize $g(x)$ using standard differential calculus or inequality methods, determine the quadratic vertex coordinates, and deduce the boundary maximum on $[1, 2]$[cite: 113, 116].

§2 Advanced Descriptive Problems

Problem 6

Let $P(x) = x^4 - 8x^2 + 21$ and $Q(y) = y^2 - 6y + 10$ be real polynomials where $x, y \in \mathbb{R}$. Given that $P(x) \cdot Q(y) = 5$, find all possible values of the sum $x + y$.
Source: **Math Vanguard Mock Test [cite: 6] (UGB – Q1 [cite: 136])**

Answer Focus: Re-architect each polynomial expression by completing the squares to ascertain global minimum bounds for $P(x)$ and $Q(y)$[cite: 137].

Problem 7

Analyze the number of real solutions to the absolute value algebraic equation:

$$|x + 1| + |x + 2| + |x + 3| = a$$

for $x \in [-4, 4]$ based on the real parameter a. Determine all possible counts of solutions this system can possess.
Source: **Math Vanguard Mock Test [cite: 6] (UGB – Q2 [cite: 150])**

Answer Focus: Define the absolute value expression as a piecewise continuous map to construct its corresponding graph, mapping the horizontal intersection possibilities[cite: 151].

Problem 8

Let α and β be the roots of the quadratic equation $x^2 - kx + 1 = 0$ (where $k > 2$). Determine all possible simplified algebraic or trigonometric expressions for the total value of:

$$\sin(\alpha^2 + 1) + \sin\left(\frac{k^2}{1+\beta^2}\right)$$

Source: **Math Vanguard Mock Test [cite: 6] (UGB – Q3 [cite: 156])**

Answer Focus: Exploit Vieta's relationships ($\alpha\beta = 1$ and $\alpha + \beta = k$) together with root substitutions to simplify the inner trigonometric fractional components[cite: 157, 160].

Problem 9

Find all real values of the parameter p for which the inequality

$$\log_{(p^2+1)}(3x^2 - 2x - p + 6) \geq \cos^2\theta(1 - 4\sin^2\theta)^2 - \cos^2 3\theta$$

holds true for all real values of x and θ.

Source: **Math Vanguard Mock Test [cite: 6] (UGB – Q4 [cite: 165])**

Answer Focus: Decouple the inequality by showing that the trigonometric right-hand side simplifies to a static constant or global maximum, then optimize the quadratic argument under a shifting log base parameter[cite: 165].

Problem 10

Consider the parametric quadratic function $f(x) = x^2 - (b+1)x + 2(a-5)$ where $a, b \in \mathbb{N}$.

a) If the equation $f(x) = 0$ has two distinct real roots for all real values of b, find the sum of all possible values of a.

b) If the inequality $f(-1) > 0$ holds for all possible valid values of a, determine the least possible value of b.

Source: **Math Vanguard Mock Test [cite: 6] (UGB – Q5 [cite: 165])**

Answer Focus: Determine structural constraints using the quadratic discriminant $\Delta > 0$ as an inequality across all real parameters, alongside evaluating functional boundaries under standard natural number sets[cite: 165, 166].